

06_Class_Activity

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In class activity 6:

What did we do last time in activity 5?

- Understanding standard normal distributions and z-scores
- Calculating and interpreting standard error
- Creating confidence intervals
- Working with the Student's t-distribution

Today's focus:

- Review more r code
- understand α alpha and β beta errors
- do more
 - 1 sample t tests
 - 2 sample t tests

Goes with Lecture 6

```
# Install packages if needed (uncomment if necessary)
# install.packages("readr")
# install.packages("tidyverse")
# install.packages("car")
# install.packages("here")

# Load libraries
library(patchwork)
library(car)          # For diagnostic tests
library(tidyverse)    # For data manipulation and visualization
library(readxl)

# Load the pine needle data
# Use here() function to specify the path
pine_switch_df <- read_excel("data/class_pine needle length switched.xlsx")

# Examine the first few rows
head(pine_switch_df)
```

```
# A tibble: 6 × 5
  group tree_no tree_char side length_mm
<chr>   <dbl> <chr>    <chr>    <dbl>
1 five     1 tree_1 sunny     22.7
2 five     1 tree_1 sunny     21.1
3 five     1 tree_1 sunny     18.6
4 five     1 tree_1 sunny     18.6
5 five     1 tree_1 sunny     21.0
6 five     1 tree_1 sunny     18.9
```

```
ps_df <- pine_switch_df %>%
  group_by(group, tree_no, tree_char, side) %>%
  summarise(length_mm = mean(length_mm, na.rm=TRUE))
```

`summarise()` has grouped output by 'group', 'tree_no', 'tree_char'. You can override using the `.groups` argument.

```
ps_shady_df <- ps_df %>%
  filter(side == "shady")

ps_sunny_df <- ps_df %>%
  filter(side == "sunny")
```

```
head(ps_df)
```

```
# A tibble: 6 × 5
# Groups:   group, tree_no, tree_char [3]
  group          tree_no tree_char side length_mm
<chr>          <dbl> <chr>   <chr>    <dbl>
1 big_fat_fecund_female_fish      2 tree_2  shady     15.4
2 big_fat_fecund_female_fish      2 tree_2  sunny     13.2
3 bill                      3 tree_3  shady     16.7
4 bill                      3 tree_3  sunny     16.0
5 ciabatta                   5 tree_5  shady     19.1
6 ciabatta                   5 tree_5  sunny     17.7
```

Part 1: Exploratory Data Analysis

Before conducting hypothesis tests, we should always explore our data to understand its characteristics.

Let's calculate summary statistics and create visualizations.

Activity: Calculate basic summary statistics for pine needle length

```
# YOUR TASK: Calculate summary statistics for pine needle length
# Hint: Use summarize() function to calculate mean, sd, n, etc.

# Create a summary table for all pine needles
pine_summary <- ps_df %>%
  group_by(side) %>%
  summarize(
    mean_length = mean(length_mm, na.rm=TRUE),
    sd_length = sd(length_mm, na.rm=TRUE),
    n = sum(!is.na(length_mm)),
    se_length = sd_length / (n^0.5),
    t_critical = qt(0.975, df = n - 1), # 95% CI uses 0.975 (two-tailed)
    ci_lower = mean_length - t_critical * se_length,
    ci_upper = mean_length + t_critical * se_length
  )

print(pine_summary)
```

```
# A tibble: 2 × 8
  side mean_length sd_length n se_length t_critical ci_lower ci_upper
<chr>    <dbl>    <dbl> <int>    <dbl>    <dbl>    <dbl>    <dbl>
1 shady      17.6      2.51     8     0.886      2.36     15.5     19.7
2 sunny      16.2      2.64     8     0.934      2.36     13.9     18.4
```

```
# Now calculate summary statistics by wind exposure
# YOUR CODE HERE
```

Part 1: Visualizing the Data

Activity: Create visualizations of pine needle length

Create a histogram and a boxplot to visualize the distribution of pine needle length values.

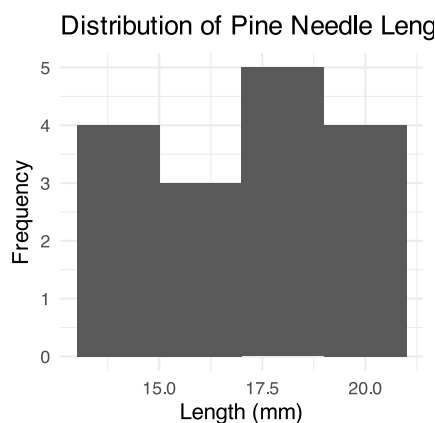
Effective data visualization helps us understand:

- The central tendency
- The spread of the data
- Potential outliers
- Shape of distribution

Your Task

```
# YOUR TASK: Create a histogram of pine needle length
# Hint: Use ggplot() and geom_histogram()

# Histogram of all pine needle lengths
ggplot(ps_df, aes(x = length_mm)) +
  geom_histogram(binwidth = 2) +
  labs(title = "Distribution of Pine Needle Lengths",
       x = "Length (mm)",
       y = "Frequency") +
  theme_minimal()
```



```
# how can you do this by side to see both plots
```

```
# Boxplot of pine needle length by sun exposure
# YOUR CODE HERE
```

what is the Effect size or difference in means?

💡 Practice Exercise: Calculate Effect size

We could also look at the difference in means... some cool code here

```
# Assuming your dataframe is called df
pine_summary %>%
  summarize(difference = mean_length[side == "sunny"] - mean_length[side == "shady"])
```

```
# A tibble: 1 × 1
  difference
    <dbl>
1      -1.45
```

Part 1: Two Sample T-Test

Now, let's compare pine needle lengths between windward and leeward sides of trees.

Question: **Is there a significant difference in needle length between the windward and leeward sides?**

This requires a two-sample t-test.

Two-sample t-test compares means from two independent groups.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where:

- $\bar{x}_1$ and $\bar{x}_2$: These represent the sample means of the two groups you're comparing
- s_p^2 : This is the pooled variance, calculated as: $s_p^2 = [(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2] / (n_1 + n_2 - 2)$, where s_1^2 and s_2^2 are the sample variances of the two groups.
- n_1 and n_2 : These are the sample sizes of the two groups.
- $\sqrt{(1/n_1 + 1/n_2)}$: This represents the pooled standard error.

Part 1: Testing Assumptions for Two-Sample T-Test

Activity: Test assumptions for two-sample t-test

For a two-sample t-test, we need to check:

1. Normality within each group
2. Equal variances between groups (for standard t-test)
3. Independent observations

If assumptions are violated:

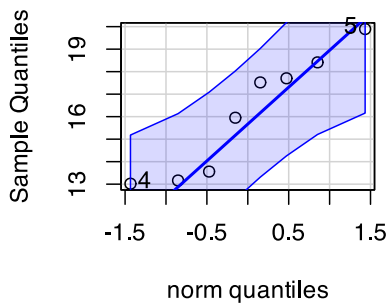
- Welch's t-test (unequal variances)
- Non-parametric alternatives (Mann-Whitney U test)

Your task

```
# YOUR TASK: Test normality of sunny pine needle lengths
# QQ Plot
qqPlot(ps_sunny_df$length_mm,
```

```
main = "QQ Plot for Windward Pine Needles",
ylab = "Sample Quantiles")
```

QQ Plot for Windward Pine Nee



```
[1] 5 4
```

```
# Testing normality for each group
# Sunny group

shapiro_lee_sunny<- shapiro.test(ps_sunny_df$length_mm)

print(shapiro_lee_sunny)
```

Shapiro-Wilk normality test

```
data:  ps_sunny_df$length_mm
W = 0.89994, p-value = 0.2886
```

shady group

```
# Sunny side group
# YOUR CODE HERE for shady group normality test
# Sunny group

shapiro_lee_shady <- shapiro.test(ps_shady_df$length_mm)

print(shapiro_lee_shady)
```

Shapiro-Wilk normality test

```
data:  ps_shady_df$length_mm
W = 0.96639, p-value = 0.8683
```

Remember you can always do it in one go

```
# there are always two ways
# Test for normality using Shapiro-Wilk test for each side group
# All in one pipeline using tidyverse approach
normality_results_both <- ps_df %>%
  group_by(side) %>%
  summarize(
    shapiro_stat = shapiro.test(length_mm)$statistic,
    shapiro_p_value = shapiro.test(length_mm)$p.value,
    normal_distribution = if_else(shapiro_p_value > 0.05, "Normal", "Non-normal")
  )

# Print the results
print(normality_results_both)
```

```
# A tibble: 2 × 4
  side shapiro_stat shapiro_p_value normal_distribution
<chr>    <dbl>         <dbl> <chr>
1 shady    0.966         0.868 Normal
2 sunny    0.900         0.289 Normal
```

Conduct a Levene's Test

```
# Test for equal variances
# YOUR TASK: Conduct Levene's test for equality of variances
levene_test <- leveneTest(length_mm ~ side, data = ps_df)
```

Warning in leveneTest.default(y = y, group = group, ...): group coerced to factor.

```
print(levene_test)
```

```
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  1  0.2062 0.6567
      14
```

Part 2: Conducting the Two-Sample T-Test

Activity: Conduct a two-sample t-test

Now we can compare the mean pine needle lengths between sunny and shady sides.

$H_0: \mu_1 = \mu_2$ (The mean needle lengths are equal)

$H_1: \mu_1 \neq \mu_2$ (The mean needle lengths are different)

Deciding between:

- Standard t-test (equal variances)
- Welch's t-test (unequal variances)

Based on our Levene's test result.

```
# YOUR TASK: Conduct a two-sample t-test
# Use var.equal=TRUE for standard t-test or var.equal=FALSE for Welch's t-test

# Standard t-test (if variances are equal)
t_test_result <- t.test(length_mm ~ side, data = ps_df, var.equal = TRUE)
print("Standard two-sample t-test:")
```

```
[1] "Standard two-sample t-test:"
```

```
print(t_test_result)
```

Two Sample t-test

```
data: length_mm by side
t = 1.1279, df = 14, p-value = 0.2783
alternative hypothesis: true difference in means between group shady and group sunny is not
equal to 0
95 percent confidence interval:
 -1.309330  4.214005
sample estimates:
mean in group shady mean in group sunny
      17.60561      16.15328
```

```
# Calculate t-statistic manually (optional - uggg - maybe )
# YOUR CODE HERE: t = (mean1 - mean2) / sqrt((s1^2/n1) + (s2^2/n2))
```

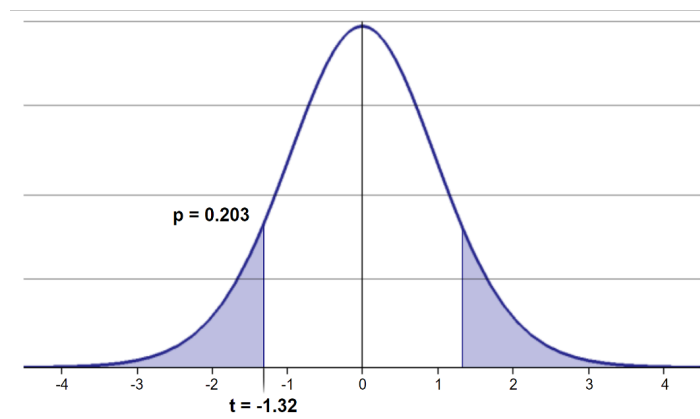
Part 2: Interpreting and Reporting Two-Sample T-Test Results

Activity: Interpret the results of the two-sample t-test

What can we conclude about the needle lengths on windward vs. leeward sides?

How to report this result in a scientific paper:

“A two-tailed, two-sample t-test at $\alpha=0.05$ showed [a significant/no significant] difference in needle length between windward ($M = \dots$, $SD = \dots$) and leeward ($M = \dots$, $SD = \dots$) sides of pine trees, $t(\dots) = \dots$, $p = \dots$ ”



What is Power

Statistical power represents the probability of detecting a true effect (rejecting the null hypothesis when it is false). In this case, with a power of 97%, there's a 97% chance of detecting a true difference of 30 units between the means of the two groups if such a difference actually exists.

A power analysis like this is typically done for one of these purposes:

1. Before data collection to determine required sample size
2. After a study to evaluate if the sample size was adequate
3. To determine the minimum detectable effect size with the given sample

With 97% power, this test has excellent ability to detect the specified effect size. Generally, **80% power is considered acceptable**, so 97% indicates a very well-powered study for detecting a difference of 30mm between the groups.

```
# Calculate power for detecting a 1 mm difference
side_diff <- 1

# Get sample sizes
sunny_n <- nrow(ps_sunny_df)
shady_n <- nrow(ps_shady_df)

# Calculate pooled standard deviation (fixed syntax)
sun_sd_pooled <- sqrt((var(ps_sunny_df$length_mm) * (sunny_n - 1) +
                      var(ps_shady_df$length_mm) * (shady_n - 1)) /
                      (sunny_n + shady_n - 2))

# Calculate Cohen's d effect size
sun_effect_size <- side_diff / sun_sd_pooled

# Calculate degrees of freedom
sun_df <- sunny_n + shady_n - 2

# Set significance level
sun_alpha <- 0.05

# Calculate power (fixed parameters)
sun_power <- power.t.test(n = min(sunny_n, shady_n),
                          delta = side_diff, # Raw difference, not effect size
                          sd = sun_sd_pooled, # Use pooled SD, not side_diff
                          sig.level = sun_alpha, # Use 0.05, not 0.5
                          type = "two.sample",
                          alternative = "two.sided")

# Display results
print("Sample sizes:")
```

```
[1] "Sample sizes:"
```

```
print(paste("Sunny:", sunny_n, "Shady:", shady_n))
```

```
[1] "Sunny: 8 Shady: 8"
```



```
print(paste("Pooled SD:", round(sun_sd_pooled, 3)))
```

```
[1] "Pooled SD: 2.575"
```

```
print(paste("Effect size (Cohen's d):", round(sun_effect_size, 3)))
```

```
[1] "Effect size (Cohen's d): 0.388"
```

```
print("")
```

```
[1] ""
```

```
print("Power analysis results:")
```

```
[1] "Power analysis results:"
```

```
print(sun_power)
```

Two-sample t test power calculation

```
      n = 8
  delta = 1
     sd = 2.575237
sig.level = 0.05
  power = 0.1083557
alternative = two.sided
```

NOTE: n is number in *each* group

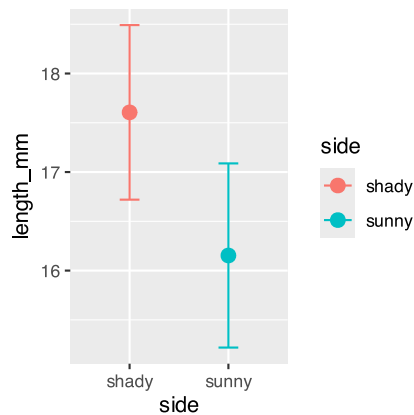
Now to make a final plot

Typically we will make a plot that has the mean and standard error on it to represent the data

your Task is to make this plot

```
pine_mean_se <- ps_df %>%
  ggplot(aes(side, length_mm, color = side))+
  stat_summary(fun = "mean", na.rm=TRUE, geom="point", size = 3)+
  stat_summary(fun.data = "mean_se", width = 0.2, geom = "errorbar")

pine_mean_se
```



Summary and Conclusions

In this activity, we've:

1. Formulated hypotheses about pine needle length
2. Tested assumptions for parametric tests
3. Conducted a two-sample t-tests
4. Visualized data using appropriate methods
5. Learned how to interpret and report t-test results

Key takeaways:

- Always check assumptions before conducting tests
- Visualize your data to understand patterns
- Report results comprehensively
- Consider alternatives when assumptions are violated

Reflection Questions

After completing the activities, discuss these questions with your group:

1. How does sample size affect our confidence in estimating the population mean?
2. Why is the t-distribution more appropriate than the normal distribution when working with small samples?
3. When comparing two populations, what can we learn from looking at confidence intervals versus performing a t-test?
4. How would you explain the concept of statistical significance to someone who has never taken a statistics course?
5. What do we do if assumptions FAIL!!!